KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3112 | 3 Credits: 4 | Program Core | Source-grounded

Production Engineering II

 To introduce the basic production processes and technologies of relevance to the manufacturing industry and related sectors

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3112 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3112.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Tool and Die Engineering
Course code	3112
Course title	Production Engineering II
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the basic production processes and technologies of relevance to the manufacturing industry and related sectors
- To impart knowledge on detailed machining aspects so as to select, operate and
- control the appropriate machining method and surface finishing processes for specific applications

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying on lathe and drilling machine Describe the functions and machining operations of	See official syllabus
CO2	15 Understanding Planer, Shaper and Slotter Interpret various operations of milling machines and	See official syllabus
CO3	13 Applying gear generating processes Illustrate abrasive processes and various types of	See official syllabus
CO4	15 Applying operation on grinding machines	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 2
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	drilling machine	4	As specified
Module 2	Slotter	4	As specified
Module 3	processes	4	As specified
Module 4	machines	4	As specified

MODULE 1

drilling machine

This module is presented from the official syllabus scope for Production Engineering II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: drilling machine
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Applying Lathe and its functions

4 Applying Lathe and its functions is an official topic in Module 1 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying Lathe and its functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying lathe and its functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying Lathe and its functions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 4 Applying Lathe and its functions [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To demonstrate different Lathe operations

To demonstrate different Lathe operations is an official topic in Module 1 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To demonstrate different Lathe operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to demonstrate different lathe operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To demonstrate different Lathe operations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] To demonstrate different Lathe operations [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To describe the working of Drilling machine

To describe the working of Drilling machine is an official topic in Module 1 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To describe the working of Drilling machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe the working of drilling machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To describe the working of Drilling machine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] To describe the working of Drilling machine [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and

To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and is an official topic in Module 1 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe different drilling operation 4 understanding lathe : types of lathes; basic parts and their functions; specifications ; work holding and tool holding device , operations - turning, parting off, knurling, facing, boring, drilling, threading, step turning, taper turning. drilling machines: drills - flat drills - twist drills - types of drilling machines - bench type - floor type - radial type - gang drill - multi - spindle type - principle of operation in drilling - speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. describe the functions and machining operations of planer, shaper and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling

Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
drilling machine
             To illustrate basics parts and accessories of
  M1.01                                                             4
Applying
             Lathe and its functions
  M1.02      To demonstrate different Lathe operations              4
Applying
  M1.03      To describe the working of Drilling machine            3
Understanding
  M1.04      To describe different drilling operation               4
Understanding
Contents:
Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work
holding
and tool holding device , Operations - Turning, parting off, Knurling, facing,
Boring,
drilling, threading, step turning, taper turning.
Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling
machines - Bench
type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle
of operation in
drilling - Speeds and feeds for various materials - drilling holes - methods of
holding drill
bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter
sinking -
counter spot facing - tapping - deep hole drilling.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Slotter

This module is presented from the official syllabus scope for Production Engineering II. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Slotter
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

To describe the working of shaper

To describe the working of shaper is an official topic in Module 2 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To describe the working of shaper using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe the working of shaper should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To describe the working of shaper means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] To describe the working of shaper [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To describe the working of Planer

To describe the working of Planer is an official topic in Module 2 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To describe the working of Planer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe the working of planer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To describe the working of Planer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ To describe the working of Planer □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

To explain the working of slotter 3 Understanding To describe quick return mechanism-crank

To explain the working of slotter 3 Understanding To describe quick return mechanism-crank is an official topic in Module 2 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To explain the working of slotter 3 Understanding To describe quick return mechanism-crank using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to explain the working of slotter 3 understanding to describe quick return mechanism-crank should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To explain the working of slotter 3 Understanding To describe quick return mechanism-crank means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ To explain the working of slotter 3 Understanding To describe quick return mechanism-crank □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating

and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating is an official topic in Module 2 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-

Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and slotted link-feed mechanism used in 4 understanding shaper, planer, and slotter shaper: types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. planer: types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation slotter: types of slotters-specifications-method of operation-whitworth quick return mechanism- feed mechanism-work holding devices-types of tools interpret various operations of milling machines and gear generating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Slotter
  M2.01      To describe the working of shaper                      4
Understanding
  M2.02      To describe the working of Planer                      4
Understanding
  M2.03      To explain the working of slotter                      3
Understanding
             To describe quick return mechanism-crank
  M2.04      and slotted link-feed mechanism used in                4
Understanding
             shaper, planer, and slotter
             Series Test - I                                        1
Contents:
Shaper: Types of shapers-specifications-standard-plain-universal principles of
operations-
drives-quick return mechanism-crank and slotted link-feed mechanism-work holding
devices-special fixture-various shaper operations.
Planer: Types of planers - description of double housing planer specifications -
principles
of operation-     drives -quick return mechanism-feed mechanism- work holding
devices
and special fixtures-types of tools and various planer operation
Slotter: Types of slotters-specifications-method of Operation-Whitworth quick
return
mechanism- feed mechanism-work holding devices-types of tools

             Interpret various operations of milling machines and gear generating
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

processes

This module is presented from the official syllabus scope for Production Engineering II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: processes
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Applying of milling machines To describe the work, tool holding devices

4 Applying of milling machines To describe the work, tool holding devices is an official topic in Module 3 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying of milling machines To describe the work, tool holding devices using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying of milling machines to describe the work, tool holding devices should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying of milling machines To describe the work, tool holding devices means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Applying of milling machines To describe the work, tool holding devices [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding and milling cutters To demonstrate milling process and milling

3 Understanding and milling cutters To demonstrate milling process and milling is an official topic in Module 3 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and milling cutters To demonstrate milling process and milling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and milling cutters to demonstrate milling process and milling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and milling cutters To demonstrate milling process and milling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding and milling cutters To demonstrate milling process and milling □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

3 Applying operations

3 Applying operations is an official topic in Module 3 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying operations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Applying operations □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding

To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding is an official topic in Module 3 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to explain gear generating processes 3 understanding milling machines: types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-t-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. gear generating processes: gear milling-gear hobbing--gear finishing processes- burnishing-shaving-grinding and lapping; gear materials-cast iron, steel, alloy steels, brass, bronze, aluminum and nylon illustrate abrasive processes and various types of operation on grinding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes-Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
processes
             To demonstrate the functions and operations
  M3.01                                                             4
Applying
             of milling machines
             To describe the work, tool holding devices
  M3.02                                                             3
Understanding
             and milling cutters
             To demonstrate milling process and milling
  M3.03                                                             3
Applying
             operations

  M3.04      To explain Gear Generating Processes                   3
Understanding

Contents:
Milling Machines: Types-column and knee type-plain- universal milling machine-
vertical
milling machine-specification of milling machines principles of operation-work
and tool
holding devices- arbor-stub arbor spring collet-adapter-milling cutters-
cylindrical milling
cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling
cutter-woodruff
milling      cutter-fly   cutter-nomenclature      of    cylindricalmilling
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

machines

This module is presented from the official syllabus scope for Production Engineering II. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: machines
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

To describe the aspects of abrasive processes

To describe the aspects of abrasive processes is an official topic in Module 4 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To describe the aspects of abrasive processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe the aspects of abrasive processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To describe the aspects of abrasive processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] To describe the aspects of abrasive processes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To demonstrate various grinding operations.

To demonstrate various grinding operations. is an official topic in Module 4 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To demonstrate various grinding operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to demonstrate various grinding operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To demonstrate various grinding operations. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] To demonstrate various grinding operations. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

To Illustrate the selection of grinding wheel

To Illustrate the selection of grinding wheel is an official topic in Module 4 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify To Illustrate the selection of grinding wheel using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to illustrate the selection of grinding wheel should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What To Illustrate the selection of grinding wheel means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ To Illustrate the selection of grinding wheel □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/

Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/ is an official topic in Module 4 of Production Engineering II. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the micro finishing operations 4 understanding abrasive processes : types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels - dressing and truing of wheels - balancing of grinding wheels. finishing: micro finishing operations - honing, lapping, superfinishing. text/ reference: t/r book title/author introduction of basic manufacturing processes and workshop technology, t1 rajendersingh, new age international (p) ltd. new delhi- 110002, 2006 elements of workshop technology volume i & ii, hajrachowdry& bhatt r1 acharaya, media promoters, 11th edition, 2007 r2 manufacturing process begeman, tata mcgraw hill, new delhi. r3 production technology –h.m.t,khanna publishers. r4 a text book of production engineering – p.c. sharma, s.chand& co. online resources: sl.no website link 1 http://www.mit.edu/ 2 https://nptel.ac.in/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
machines

  M4.01      To describe the aspects of abrasive processes         3
Understanding

  M4.02      To demonstrate various grinding operations.           5
Applying

  M4.03      To Illustrate the selection of grinding wheel         3
Applying

  M4.04      Explain the micro finishing operations                4
Understanding

             Series Test – II                                      1
 Contents:
 Abrasive Processes : Types and classification - specifications - rough grinding
- pedestal
 grinders - portable grinders - belt grinders - precision grinding cylindrical
grinder -
 centerless grinders - surface grinder - tool and cutter grinder - planetory
grinders -
 principles of operations - grinding wheels abrasives - natural and artificial
diamond wheels
 - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes
- standard
 marking systems of grinding wheels - selection of grinding wheel - mounting of
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 4 Applying Lathe and its functions. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying Lathe and its functions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain To demonstrate different Lathe operations. (3 marks)

Answer: Begin with the standard definition or identity of *To demonstrate different Lathe operations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain To describe the working of Drilling machine. (3 marks)

Answer: Begin with the standard definition or identity of *To describe the working of Drilling machine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep

hole drilling. Describe the functions and machining operations of Planer, Shaper and. (3 marks)

Answer: Begin with the standard definition or identity of *To describe different drilling operation 4 Understanding Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling. Describe the functions and machining operations of Planer, Shaper and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain To describe the working of shaper. (3 marks)

Answer: Begin with the standard definition or identity of *To describe the working of shaper*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain To describe the working of Planer. (3 marks)

Answer: Begin with the standard definition or identity of *To describe the working of Planer*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain To explain the working of slotter 3 Understanding To describe quick return mechanism-crank. (3 marks)

Answer: Begin with the standard definition or identity of *To explain the working of slotter 3 Understanding To describe quick return mechanism-crank*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating. (3 marks)

Answer: Begin with the standard definition or identity of *and slotted link-feed mechanism used in 4 Understanding shaper, planer, and slotter Shaper: Types of shapers-specifications-standard-plain-universal principles of operations- drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations. Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools Interpret various operations of milling machines and gear generating*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 4 Applying of milling machines To describe the work, tool holding devices. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying of milling machines To describe the work, tool holding devices*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding and milling cutters To demonstrate milling process and milling. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding and milling cutters To demonstrate milling process and milling*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying operations. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying operations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding. (3 marks)

Answer: Begin with the standard definition or identity of *To explain Gear Generating Processes 3 Understanding Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment. Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes- Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon Illustrate abrasive processes and various types of operation on grinding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain To describe the aspects of abrasive processes. (3 marks)

Answer: Begin with the standard definition or identity of *To describe the aspects of abrasive processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain To demonstrate various grinding operations.. (3 marks)

Answer: Begin with the standard definition or identity of *To demonstrate various grinding operations.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain To Illustrate the selection of grinding wheel. (3 marks)

Answer: Begin with the standard definition or identity of *To Illustrate the selection of grinding wheel*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing:

micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the micro finishing operations 4 Understanding Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetory grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels -Dressing and Truing of wheels - Balancing of grinding wheels. Finishing: micro finishing operations - honing, lapping, superfinishing. Text/ Reference: T/R Book Title/Author Introduction of Basic Manufacturing Processes and Workshop Technology, T1 Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006 Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt R1 Acharaya, Media Promoters, 11th Edition, 2007 R2 Manufacturing Process Begeman, Tata McGraw Hill, New Delhi. R3 Production Technology –H.M.T,Khanna Publishers. R4 A Text Book of Production engineering – P.C. Sharma, S.Chand& Co. Online Resources: Sl.No Website Link 1 http://www.mit.edu/ 2 https://nptel.ac.in/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Applying Lathe and its functions**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **To describe the working of shaper**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Applying of milling machines To describe the work, tool holding devices**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **To describe the aspects of abrasive processes**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link http://www.mit.edu/ https://nptel.ac.in/

IN-PAGE SEARCH

Search this lesson